Applied Econometrics: Session 1

Foundations of Causal Inference

PhD Intensive Course

Lecture 1

April 1, 2026

How These Slides Are Structured

- This deck follows the lecture sequence: "Questions about Questions" and "The Experimental Ideal".
- Every core idea appears in three layers: question, intuition, and formula.
- The target is self-study: students should be able to revise directly from these slides.
- Detailed algebra is in appendix with back-and-forth buttons.

Roadmap for Today

1. Questions About Questions

2. Synthesis for Lecture 1

Why Start with Questions, Not Methods?

- Good econometrics cannot rescue a weak research question.
- A coherent research agenda is the first identification device.
- Experimental way of thinking before choosing estimators.

Core message

First define the causal question and ideal experiment; only then choose empirical strategy.

The Four Research FAQs

1. What is the causal relationship of interest?
2. What is the ideal experiment that answers it?
3. What is the identification strategy in available data?
4. What is the mode of statistical inference?

These are sequential, not optional.

FAQ 1: What Is the Causal Relationship?

Example: effect of schooling on wages.

- Descriptive statement: high-school and college graduates earn differently.
- Causal statement: what would happen to the same person's wage if schooling changed?
- Policy relevance: tuition subsidies, compulsory schooling laws, financial aid.

Practical rule

If the question has no counterfactual interpretation, causal methods are not yet the right tool.

FAQ 2: What Is the Ideal Experiment?

- Imagine an RCT with no budget and no practical constraints.
- This clarifies what must vary and what must stay fixed.
- Even hypothetical experiments improve question quality.

Example:

- To estimate schooling returns, ideal design randomly changes years of schooling.
- Real world cannot do this perfectly, but the thought experiment defines the target.

FAQ 2 Extension: FUQ (Fundamentally Unidentified Questions)

Some questions cannot be answered by any feasible experiment. **Example:** effect of school starting age on first-grade scores.

- Older starters are older at testing (maturation effect).
- If you compare at same age, they have different time in school.
- Age and schooling exposure are mechanically linked in child samples.

Lesson

Sometimes data cannot identify the target because the target itself is not separable.

Appendix details

FAQ 3: What Is the Identification Strategy?

- Identification strategy means: how observational data mimic an experiment.
- Typical sources: policy rules, institutional thresholds, lotteries, timing differences.
- Strategy must be described before estimation details.

Guiding principle: design first, mechanics second.

FAQ 4: What Is the Mode of Inference?

- What population does the sample represent?
- What is the sampling/assignment process?
- How should standard errors be computed (heteroskedasticity, clustering, design effects)?

Why this matters

You can have a strong design but still get misleading precision if inference is wrong.

Workflow for Applied Work

1. Write the causal estimand in words and symbols.
2. Describe an ideal randomized trial.
3. State what real-world variation approximates that trial.
4. Explain why inference assumptions match the data structure.

This sequence prevents most beginner mistakes.

The Experimental Ideal: Motivation

- Random assignment is the benchmark for causal credibility.
- Nonexperimental comparisons often point in the wrong direction.
- We need to understand why selection bias appears and how experiments remove it.

Hospital Example

Question: Do hospitals make people healthier?

- Naive data comparison often shows hospitalized people in worse health.
- This does not prove hospital care is harmful.
- Sicker people are more likely to go to hospital in the first place.

Interpretation

Observed differences combine treatment effects and pre-treatment differences.

Potential Outcomes Notation

For each unit i:

$$Y_{1i} = \text{outcome if treated}, \qquad Y_{0i} = \text{outcome if untreated}.$$

Treatment indicator: $D_i \in \{0, 1\}$. Observed outcome:

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i.$$

- Individual causal effect: $Y_{1i} - Y_{0i}$.
- We never observe both potential outcomes for the same unit at the same time.

Fundamental Problem of Causal Inference

- For treated units, Y_{0i} is missing.
- For untreated units, Y_{1i} is missing.
- Therefore causal inference is a missing-counterfactual problem.

Key consequence

We need assumptions or design features to reconstruct missing counterfactual means.

Selection Bias Decomposition

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}.$$

- Left side is observed mean difference.
- First term is causal effect for treated units.
- Second term is non-causal baseline difference between groups.

Appendix details

Numeric Intuition for Selection Bias

Suppose observed difference is -0.70 health points (higher = worse health).

- Treated group is sicker even without treatment: selection term is -1.00.
- Then treatment effect on treated is:

$$\text{ATT} = -0.70 - (-1.00) = +0.30.$$

Interpretation

Naive comparison says treatment looks harmful; corrected effect is beneficial.

Why Random Assignment Solves the Selection Problem

Random assignment gives:

$$D_i \perp\!\!\!\perp (Y_{1i}, Y_{0i}).$$

Therefore:

$$\mathbb{E}[Y_{0i} \mid D_i = 1] = \mathbb{E}[Y_{0i} \mid D_i = 0]$$

so selection bias term is zero.

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i}].$$

- Experiments create comparability by design, not by modeling assumptions.

Appendix details

What Experiments Do Not Automatically Solve

- Attrition can destroy comparability if dropout differs by treatment status.
- Noncompliance weakens treatment contrast (assignment vs receipt).
- Spillovers can violate stable treatment definitions.
- Measurement problems can still bias results.

Conclusion: randomization is powerful, but implementation details still matter.

Case Study: Tennessee STAR

- Large education RCT on class size in early grades.
- Students assigned to small class, regular class, or regular class plus aide.
- Goal: estimate causal effect of class size on achievement.

Why influential:

- Design is transparent.
- Policy relevance is immediate.
- Results differ from many naive observational comparisons.

STAR Balance and Interpretation

Typical first check in any RCT: covariate balance across groups.

- Race, age, and free-lunch status should be similar if randomization worked.
- Class-size differences should be large across arms (first-stage check).
- Attrition differences need attention because they can induce bias later.

Empirical lesson

Even in RCTs, credibility comes from both assignment and execution quality.

Experimental Effects from Means and Regressions

In a two-group experiment, effect estimate is difference in means:

$$\hat{\tau} = \bar{Y}_{D=1} - \bar{Y}_{D=0}.$$

This is numerically equivalent to OLS coefficient from

$$Y_i = \alpha + \rho D_i + u_i.$$

- Here $\hat{\rho} = \hat{\tau}$.
- Regression is a convenient language for experimental analysis.

Appendix details

Regression Analysis of Experiments

With constant treatment effect model:

$$Y_i = \alpha + \rho D_i + \eta_i,$$

where ρ is causal effect and η_i captures untreated heterogeneity.

- In randomized data, D_i is orthogonal to η_i.
- OLS recovers causal ρ without omitted variable bias.
- Adding pre-treatment controls can improve precision.

Appendix details

From Ideal Experiments to Quasi-Experiments

- True RCTs are often expensive, slow, or infeasible.
- Strategy: search for natural variation that is "as good as random".
- Benchmark remains the notional randomized trial.

Example:

- Class-size caps create discontinuous class-size changes around enrollment thresholds.
- Units near threshold can be highly comparable.

Chapter 2 Practical Checklist

1. Define treatment and outcome precisely.
2. Write potential outcomes and target estimand.
3. Decompose observed difference into effect plus selection.
4. Ask if assignment is random or plausibly as-if random.
5. Audit threats: attrition, noncompliance, spillovers, inference.

Integrated Summary of Chapters 1 and 2

- Chapter 1 gives the research-design blueprint (four FAQs).
- Chapter 2 gives the formal causal engine (potential outcomes + selection problem + random assignment).
- Regression enters as a practical estimator for experimental contrasts.
- Credible empirical work combines clear question, design logic, and valid inference.

What Students Should Be Able to Do After This Lecture

- Distinguish causal questions from descriptive questions.
- Write and interpret the selection-bias decomposition.
- Explain why random assignment identifies average treatment effects.
- Understand why OLS with a treatment dummy equals difference in means in experiments.
- Use the four FAQs to structure research proposals.

Appendix Map

- A1: Selection-bias decomposition proof.
- A2: Random assignment identification proof.
- A3: OLS equals difference in means derivation.
- A4: Constant-effect regression and causal interpretation.
- A5: FUQ school-starting-age algebra.

Appendix A1: Selection-Bias Decomposition Proof

Start with observed gap:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0].$$

$$\mathbb{E}[Y_i \mid D_i = 1] = \mathbb{E}[Y_{1i} \mid D_i = 1], \quad \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{0i} \mid D_i = 0].$$

Add and subtract $\mathbb{E}[Y_{0i} \mid D_i = 1]$:

$$\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] + (\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]).$$

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Appendix A2: Why Random Assignment Identifies Causal Effects

If assignment is random:

$$D_i \perp\!\!\!\perp (Y_{1i}, Y_{0i}).$$

Then

$$\mathbb{E}[Y_{1i} \mid D_i = 1] = \mathbb{E}[Y_{1i}], \qquad \mathbb{E}[Y_{0i} \mid D_i = 0] = \mathbb{E}[Y_{0i}].$$

Therefore observed difference simplifies to

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] = \mathbb{E}[Y_{1i} - Y_{0i}].$$

Selection bias is eliminated because untreated potential outcomes are balanced by design.

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Appendix A3: OLS with Binary Treatment Equals Difference in Means

Regression:

$$Y_i = \alpha + \rho D_i + u_i, \quad D_i \in \{0, 1\}.$$

Predicted mean for controls ($D = 0$):

$$\mathbb{E}[Y \mid D = 0] = \alpha.$$

Predicted mean for treated ($D = 1$):

$$\mathbb{E}[Y \mid D = 1] = \alpha + \rho.$$

Subtract:

$$\rho = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

Sample OLS counterpart gives

$$\hat{\rho} = \bar{Y}_{D=1} - \bar{Y}_{D=0}.$$

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Appendix A4: Constant-Effect Model and Causal Interpretation

Suppose

$$Y_i = Y_{0i} + \rho D_i,$$

where ρ is constant across units. Write

$$Y_i = \alpha + \rho D_i + \eta_i,$$

with

$$\alpha = \mathbb{E}[Y_{0i}], \qquad \eta_i = Y_{0i} - \mathbb{E}[Y_{0i}].$$

If $D_i \perp\!\!\!\perp Y_{0i}$, then $\mathrm{Cov}(D_i, \eta_i) = 0$, so OLS recovers ρ. In observational data this orthogonality can fail, generating omitted-variable bias.

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Appendix A5: FUQ Algebra for School Starting Age

Let

$$\text{Current age} = \text{start age} + \text{time in school}.$$

If outcome is first-grade test score, then for children still in school:

- Varying start age changes current age at test.
- Holding current age fixed changes time in school.

So direct effect of start age is not separable from maturation and schooling exposure in this setting.

Key point

Some targets are structurally unidentified in the available comparison frame.

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